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Joint 60th Annual Northeastern/59th Annual North-Central Section Meeting - 2025

Paper No. 22-22

Presentation Time: 8:30 AM-5:30 PM

DEVELOPING A LIVE R-GIS PLATFORM FOR EARTHQUAKE MONITORING IN NEW YORK: ENHANCING PUBLIC AWARENESS OF LOW-MAGNITUDE SEISMIC EVENTS

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Low-magnitude earthquakes, while less dangerous, can occur in New York from time to time. Due to their rarity, New Yorkers are largely unfamiliar with such events, often lacking the information necessary to understand or respond effectively. The development of a live GIS-based map tailored to monitor earthquake activity exclusively within the New York region aims to address this gap.

We utilize R programming and GIS tools along with USGS earthquake data to create a user-friendly platform that informs residents about seismic activity in real time. Unlike existing global USGS earthquake maps, this initiative focuses on delivering a localized, simplified solution to meet the specific needs of the New York population. The platform ensures accessibility for all, regardless of scientific expertise, by emphasizing clarity and usability.

Designed to prevent misunderstandings and unnecessary alarm, the system minimizes the risk of seismic events being misinterpreted as terrorist actions or other disturbances—particularly during nighttime hours, when panic and distress are more likely. Through careful filtering and geospatial visualization, the platform highlights only relevant data, enhancing public awareness and fostering a more informed community response.

This dedicated GIS website bridges the gap between complex scientific data and public understanding, serving as a model for similar low-seismic regions. By improving disaster preparedness and reducing unwarranted panic, this project demonstrates the value of region-specific geohazard communication systems.

Session No. 22--Booth# 22

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Saturday, 29 March 2025: 8:30 AM-5:30 PM

North Great Hall (Bayfront Convention Center)

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